Generation of a Hidden Markov Model for the annotation of Kunitz-type domain: analysis and performance evaluation

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Abstract
Aim: The project aimed to develop a high-quality Hidden Markov Model (HMM) by curating a non-redundant training set of structural monodomains to ensure a sharp, specific statistical signal for the Kunitz family.
Results: The resulting model, achieved a near-perfect Matthews Correlation Coefficient (MCC) of 0.982 and demonstrated high discriminative power across over 570,000 UniProt entries at a consensus E-value threshold of 10^{-5}.
Keywords: Linux/Bash commands, HMM generation, MCC coefficient, Reproducible workflow
Supplementary material: https://github.com/alessiavisani/generation-of-an-Hidden-Markov-Model-for-Kunitz-type-domain

1. INTRODUCTION

Kunitz domains represent a set of active domains of proteins denoting the peculiar activity of inhibiting the function of protein degrading enzymes. This specific class of proteins is known as ***protease inhibitors***; between which the chief representative is BPTI (Bovine Pancreatic Trypsin Inhibitor), also known as aprotinin. This is considered as the "model" or "type" representative because it was the first one being extensively studied (by Moses Kunitz in the 1930s) and its structure is the gold standard for understanding how these inhibitors work. Kunitz/BPTI inhibitors are small proteins (around 6 kDa) of about 60 to 50 amino acid residues characterized by a compact structure composed of a hydrophobic core, containing a central β-sheet and three di-sulphide bridges with conserved chirality [1]. These three highly conserved di-sulphide bonds follow a specific connectivity pattern—*typically C1 -C6, C2-C4, and C3-C5* —which cross-link the polypeptide chain to create a rigid, pear-shaped framework. This covalent architecture is so robust that it allows the inhibitor to remain functional even under high temperatures or within the presence of aggressive proteolytic enzymes.

The functional heart of the Kunitz domain is the *reactive site loop*, often visualized through "P-site" nomenclature. At the center of this loop is *the residue* (often Arginine or Lysine), which acts as the primary determinant of the inhibitor's specificity.

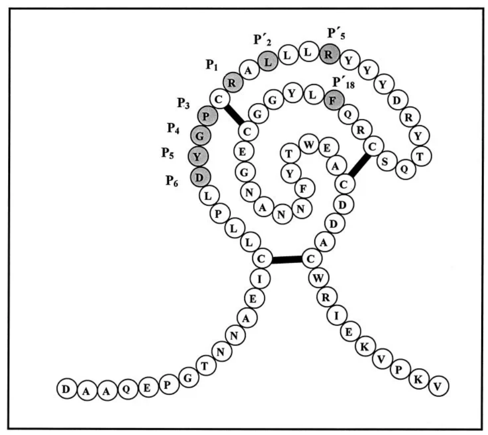

Fig 1. Model structure of the first Kunitz-type domain (KD1) of human TFPI-2. Residues mutated in this study are shaded. (Bajaj et al., 2011)

When the inhibitor encounters a target protease (like trypsin), the reactive site loop mimics the natural substrate of the enzyme. It inserts itself into the enzyme's active site like a "plug" in a socket. However, because the Kunitz domain is so tightly reinforced by its di-sulphide bridges, the enzyme is unable to cleave the inhibitor. This results in a stable, non-covalent complex that physically blocks the enzyme from degrading other proteins, effectively "turning off" the protease activity.

Beyond the foundational biochemistry, recent research has shifted from purely structural studies to therapeutic applications having high-impact on the pharmaceutical market becoming a cornerstone for drug development in viral defence, oncology and venom-based medicine.

One of the most notable recent shifts is the use of Kunitz-type inhibitors like Aprotinin (BPTI) in treating acute respiratory infections. For SARS-CoV-2 and other respiratory viruses, such as influenza, to successfully infect a human host, they must first achieve fusion with the host cell membrane. This critical entry step is not autonomous; it requires the viral "Spike" (S) protein to be chemically primed.

This activation is performed by a host cell enzyme known as TMPRSS2(Transmembrane Serine Protease 2), which cleaves the spike protein at specific sites to trigger the fusion process **[3].** Kunitz-type inhibitors, such as Aprotinin (the family's classic representative), function as potent competitive inhibitors. Aprotinin binds directly to the active site of the TMPRSS2 protease. Because the Kunitz domain is structurally rigid and reinforced by three conserved di-sulphide bridges, it occupies the enzyme's catalytic pocket without being degraded itself **[4].** By physically blocking the enzyme, the inhibitor prevents TMPRSS2 from cleaving the viral Spike protein inhibiting the essential conformational changes required to merge the viral envelope with the host cell membrane.

Given the clinical relevance of these serine protease inhibitors, the development and improvement of tools for annotating Kunitz domains are crucial. A variety of approaches has been investigated, with some relying on structural classification systems while others infer domains by clustering conserved subsequences. In this analysis, Hidden Markov Models (HMMs) have been utilized to demonstrate the effectiveness of this tool. The approach employed involved the generation and training of an HMM capable of annotating the Kunitz domain within the UniProtKB/Swiss-Prot database.

2. MATERIALS & METHODS

All materials necessary for a fully reproducible workflow, including the trained Hidden Markov Models, optimization scripts, and datasets, are publicly available at the following repository: https://github.com/alessiavisani/generation-of-an-Hidden-Markov-Model-for-Kunitz-type-domain.

2.1 Data retrieval and training set selection

The initial phase of the workflow involved selecting a representative dataset to define the essential features of the Kunitz-type protease inhibitor domain. To ensure high-quality training data, an advanced search was conducted within the RCSB PDB database, utilizing specific structural constraints to capture the domain's unique characteristics. These criteria included firstly the mandatory membership in the serine protease inhibitor family, specifically identified by the Pfam accession code PF00014; structural quality was assessed with a data collection resolution threshold of less than 3.5 Å to ensure the reliability of the atomic coordinates and finally, a strictly defined range of sequence length between 40 and 80 amino acid residues, filtering for the characteristic compact size of the Kunitz fold while excluding truncated fragments or larger multidomain proteins.

Consequently, a Custom tabular report indicating the Entry ID, Chain and sequence was downloaded (to be found in the supplementary materials). This dataset was then parsed using Linux/Bash commands to generate a refined list of unique identifiers and sequences. To remove the redundancy coming from the similarity of sequences the pdb_seqs.txt was uploaded and analyzed by the online tool MM2seqs2 filtering the search by two different thresholds: minimum sequence identity of 0.95 and minimum alignment coverage of 0.90.

MMseqs2 is a high-speed software suite designed for protein and nucleotide sequence analysis. The core strength of MMseqs2 is its ability to maintain high sensitivity while being **orders of magnitude faster** than its predecessors. It achieves this through a "k-mer" based indexing strategy and a vectorized Smith-Waterman alignment [5]. The output of MMseqs2 is a set of clusters made of different sequences; since the algorithm sorts sequences in a greedy way it picks the first available sequence and finds all other sequences that are similar enough to it; These similar sequences are "pulled" into a cluster, and the first sequence is designated the representative.

For each cluster the representative sequence identifier together with the chain of interest was extracted and put in a file of representative Ids.

2.2 Multiple Structure Alignment

The list of the representative Ids was created using the `get_chains.py` script (supplementary materials) to isolate the specific Kunitz monodomain coordinates from the source PDB files. This step was critical because PDB entries often contain multiple chains or complex heteromeric structures, whereas the target domain is consistently between 40 and 80 residues in length. By iterating through the representative list, the script extracted only the

"ATOM" records corresponding to the relevant chain ID, ensuring that the downstream alignment was focused strictly on the domain of interest. The resulting isolated chains files were then used as an input for the Multiple Structure alignment Tool mTM-align. An installation of the toolkit was preferred since some problems were encountered while using the online platform of PDBeFold; indeed mTM-align is a powerful structural alignment tool as it outperforms many other MSTA algorithms, such as Matt, MUSTANG and MultiProt. In addition, comparison with the manually curated alignments in the HOMSTRAD database shows that the automated alignments built by mTM-align are in general more accurate. Therefore, mTM-align may be used as a reliable complement for constructing multiple structure alignment for real-world applications [6].

After verifying the correctness of the MSA alignment output; the former was used as a seed alignment for the generation of the Hidden Markov Model.

2.3 Hidden Markov Model generation

Hidden Markov models (HMMs) have been extensively used in biological sequence analysis. A *hidden Markov model (HMM)* is a statistical model that can be used to describe the evolution of observable events that depend on internal factors, which are not directly observable. We call the observed event a `symbol' and the invisible factor underlying the observation a `state'. An HMM consists of two stochastic processes, namely, an invisible process of hidden states and a visible process of observable symbols. The hidden states form a *Markov chain*, and the probability distribution of the observed symbol depends on the underlying state. For this reason, an HMM is also called a doubly embedded stochastic process [7]. In our case we used the **`HMMEr`** version 3.4 (Aug 2023) by calling the command **`hmmbuild`** using as input the **`kunitz_clean.ali`** to obtain a profile HMM which have been showed to be useful for various tasks including protein classification, motif detection, and finding multiple sequence alignments [7]. Subsequentially, to visualize the quality of the constructed HMM, the Skylign tool was implied to provide a logo representation of the work performed up to this point. A logo provides a compact graphical representation of an alignment, representing each column with a stack of letters. The total height of each stack corresponds to a measure of the invariance of the column – typically, it is the *information content* of that position. The height of each letter within a stack depends on the frequency of that letter at that position [8].

2.4 Selection of the Test Set

To evaluate the discriminative power of the generated profile Hidden Markov Model (HMM), a benchmarking test set was constructed using the Swiss-Prot (reviewed) section of UniProtKB as a high-quality reference population. The validation framework relied on the establishment of two "gold standard" ground-truth datasets to assess the model's ability to distinguish between homologous and non-homologous sequences.

The **Positive Set** was defined as all reviewed sequences containing the target Kunitz domain, identified through high-confidence annotations in the Pfam database under entry **PF00014**. Conversely, the **Negative Set** served as a diverse control group representing the non-target protein space, consisting of all Swiss-Prot sequences explicitly lacking the PF00014 annotation. Utilizing the entire reviewed UniProtKB for the negative population ensures a biologically realistic distribution for calculating false discovery rates and optimizing E-value thresholds.

2.5 Model Testing

To ensure a rigorous and unbiased validation, the benchmarking test set was filtered against the sequences used for model training. A local BLAST database using **`makeblastdb`** was constructed using the UniProtKB/Swiss-Prot (reviewed) dataset as the reference. To identify exact matches between the structural templates and the test population, the PDB identifiers used in the initial mTM-align structural alignment were first converted to their corresponding UniProt accessions via the *UniProt ID Mapping* service. These representative sequences were then used as queries in a **`BLASTp`** search against the Swiss-Prot database. Any sequences in the test set exhibiting 100% identity to the training templates were discarded. This step was critical to prevent overestimation of the model's accuracy and to ensure that the validation reflects the HMM's ability to detect novel or divergent homologs rather than simply recognizing its own training data.

The benchmarking procedure was conducted by querying the profile HMM against the complete Swiss-Prot database using **`hmmsearch`** with the **--max** flag enabled. This setting disables default acceleration heuristics to ensure an exhaustive search that captures weak or divergent signals that might otherwise be bypassed. To maintain statistical consistency across datasets of varying sizes, the `-Z flag` was set to a constant value of 1000, fixing the number of comparisons used for E-value calculations. This normalization ensures that E-values remain comparable throughout the evaluation process. Because **`HMMER`** typically only reports sequences exceeding a specific probability threshold, a post-search integration step was required to maintain statistical integrity. The **`comm`** command was utilized to identify sequences from the negative database that produced no match in the initial search. These "no-hits" were re-integrated into the final dataset and assigned a high, arbitrary E-value of 100. This imputation ensures that all non-detected sequences are correctly ranked as non-members during the empirical optimization of the E-value threshold and subsequent performance plotting.

2.6 E-value optimization and MCC accountability

To pick the optimal E-value, enabling for a maximization of the classification performance, an optimization procedure was carried out on the two subsets (kunitz_set1 / kunitz_set2) derived from the random splitting and sorting of the whole original set. The performance was then tested by executing the **`performance.py`** script available in the supplementary materials, across a range of E-value thresholds from 10^{-1} to 10^{-15}. The script processed the **`hmmsearch`** results by comparing the predicted labels against the real values—encoded as 1 for the positive Kunitz class and 0 for the negative class—found in the third column of the input files.

The primary metric used to rank these thresholds was the *Matthews Correlation Coefficient (MCC).* The MCC was selected because it provides a more robust and balanced score than simple accuracy, particularly when dealing with the significant class imbalance between the limited positive sequences and the massive negative Swiss-Prot background. By identifying the E-value that maximized the MCC across the cross-validation sets, the procedure ensured the selection of a threshold that optimized the balance between sensitivity (minimizing false negatives) and specificity (minimizing false positives). Formula of the Matthews correlation coefficient can be found below. (Fig 1)

$$\mathrm{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Figure 1: Formula for the Matthews correlation coefficient

3. RESULTS

3.1 Data retrieval and training selection

The advanced search in PDB resulted in the recovery of 135 polymer entries. After the downloading of the Custom tabular report only 102 sequences were chosen according to the sequence length constraints of the Kunitz monodomain; the mentioned sequences are in fact what can be found in the **`pdb_seqs.txt`** generated. The input of the MMseqs2 was the aforementioned file; such sequences were clustered into 19 different clusters according to a minimum sequence identity of 95% aand a minimum alignment coverage of 90%. For each cluster only a representative sequence was kept and stored into the **`pdb_id.rep`** (19 sequences).

3.2 Multiple structural alignment: input and polishing

Following the generation of the initial multiple structural alignment (MSA) from the 19 selected PDB chains, a comparative analysis of sequence identity and conserved motifs was performed to verify the integrity of the training set. Visual inspection of the **`kunitz.ali`** identity matrix identified two significant structural outliers (1D0D and 5YW1) that deviated from the standard Kunitz monodomain architecture (Fig 2). These sequences exhibited a markedly low identity percentage relative to the core cluster—represented by pale regions in the heatmap—and failed to account for the six conserved cysteines essential to the Kunitz di-sulphide-bonded framework. The inclusion of these artifacts threatened to introduce substantial noise and gaps into the emission probabilities of the Hidden Markov Model.

As illustrated by the **`kunitz_clean.ali`** identity matrix, the removal of these two sequences significantly enhanced the internal consistency of the dataset. This refinement shifted the maximum sequence identity from approximately 60% in the original set to over 90% in the cleaned version, signalling a much tighter structural grouping. The resulting dataset displays a robust, homogeneous core of high-identity matches, particularly among sequences such as 3P95, 3TGI, and 3WNY, which serve as a stable statistical anchor for the subsequent HMM matching states.

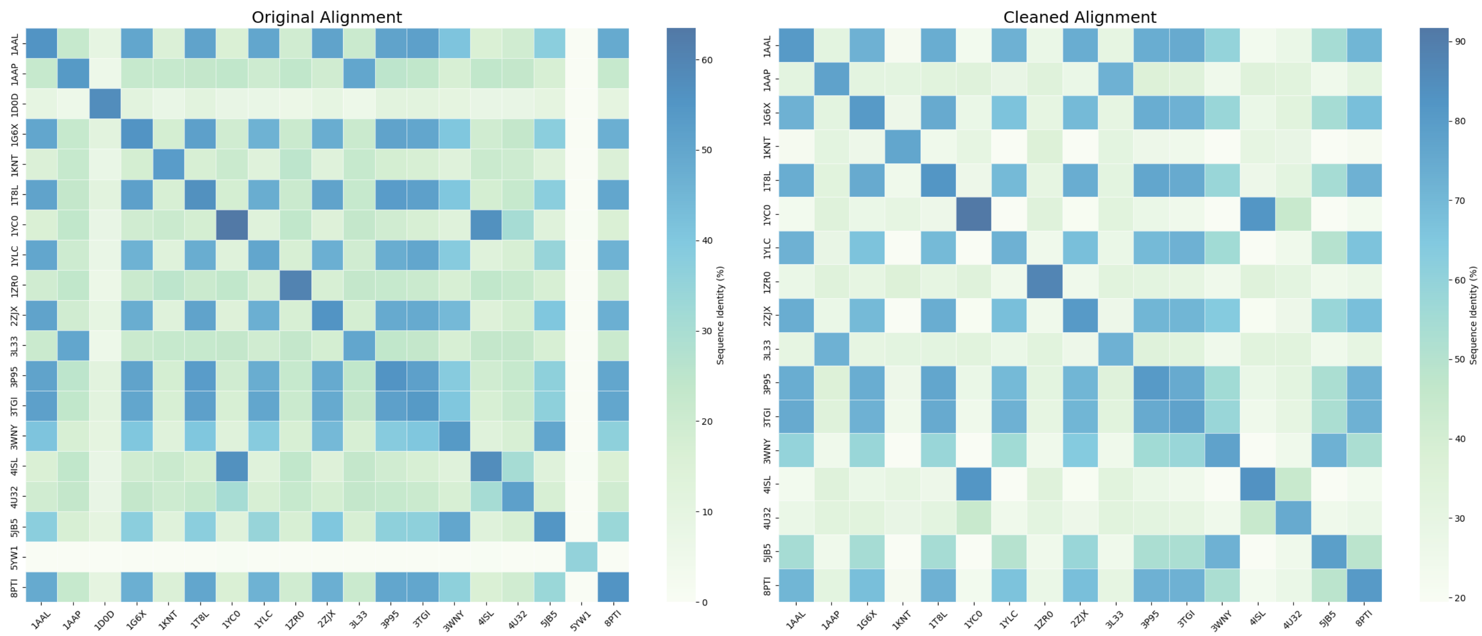

Figure 2: comparison between the original alignment (`kunitz.ali`) and the cleaned alignment (`kunitz_clean.ali`)

3.3 Hidden Markov model generation and evaluation of the quality of the model: sequence logo and emission probabilities

The Hidden Markov Model (HMM), which in input took the **`kunitz_clean.ali`**, was constructed using the **`hmmbuild`** utility from the refined structural alignment. Following the removal of structural outliers, the model was trained on a set of 17 representative sequences. The resulting HMM possessed a match state length of 57 residues, a figure that aligns closely with the biological consensus for the Kunitz monodomain.

To evaluate the quality of the model, the emission probabilities and a sequence logo were generated via Skylign, confirming the high statistical significance of the model's core features. Specifically, the emission probabilities at key match states revealed a near-absolute conservation of the six cysteine residues that form the domain's characteristic di-sulphide-bonded architecture. By utilizing a "clean" structural alignment, the model successfully minimized noise from non-representative insertions or gaps, with matching positions in the global alignment spanning from residues 10

to 69 to effectively capture the entire functional fold.

Furthermore, the sequence logo highlights that the most informative positions correspond to these structurally invariant residues, providing a robust statistical anchor for the identification of distant homologs. (Fig 3, Fig 4)

3.4 Classification and performance results

The predictive performance of the **`kunitz_clean.hmm`** was evaluated using a benchmark dataset derived from UniProtKB reviewed entries, consisting of **398 positive sequences which after the polishing of the test set were reduced to 394** (containing the Pfam PF00014 Kunitz domain) and **574,229 negative sequences**. To determine the optimal E-value threshold for classification, a 2-fold cross-validation procedure was implemented by partitioning the data into two equally sized subsets randomly partitioned and sorted.

The optimization process revealed a slight divergence in performance metrics between the two subsets. *Subset 1* achieved a perfect Matthews Correlation Coefficient (MCC) of **1.0** at an E-value threshold of 10^{-4}. Conversely, *Subset 2* reached its peak performance at a stricter threshold of 10^{-7}. This divergence indicates that *Subset 1* contains Kunitz domains with slightly lower sequence conservation, necessitating a more relaxed threshold to ensure high sensitivity. To balance sensitivity and specificity across the entire population, a consensus threshold of 10^{-5} was selected, yielding a robust MCC of approximately **0.98(** Fig 6**)**.

As illustrated in the confusion matrices for the selected threshold, the model exhibited high discriminative power (Fig5). In *Set 1*, the model identified *197 True Positives with only 1 False Positive and 0 False Negatives*. In *Set 2*, it *identified 193 True Positives with 0 False Positives, though 4 False Negatives were observed.*

These False Negatives include specific entries as **O62247** (BLI5_CAEEL), **D3GGZ8** (BLI5_HAECO), and **A0A1Q1NL17** (HA11_HYAAI), which yielded significantly weaker E-values ranging from 2.1 x 10^{-4} to 2.2. An analysis of these outliers suggests they represent extreme biological variations or distant homologs of the Kunitz fold that fall at the edge of the model's statistical confidence. Despite these cases, the model demonstrated exceptional specificity, maintaining a near-perfect classification rate against the extensive negative dataset of over 570,000 entries.

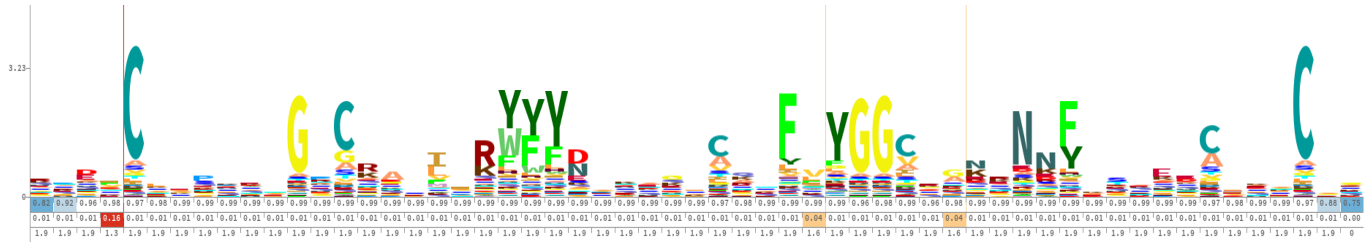

Figure 4: Skyling logo

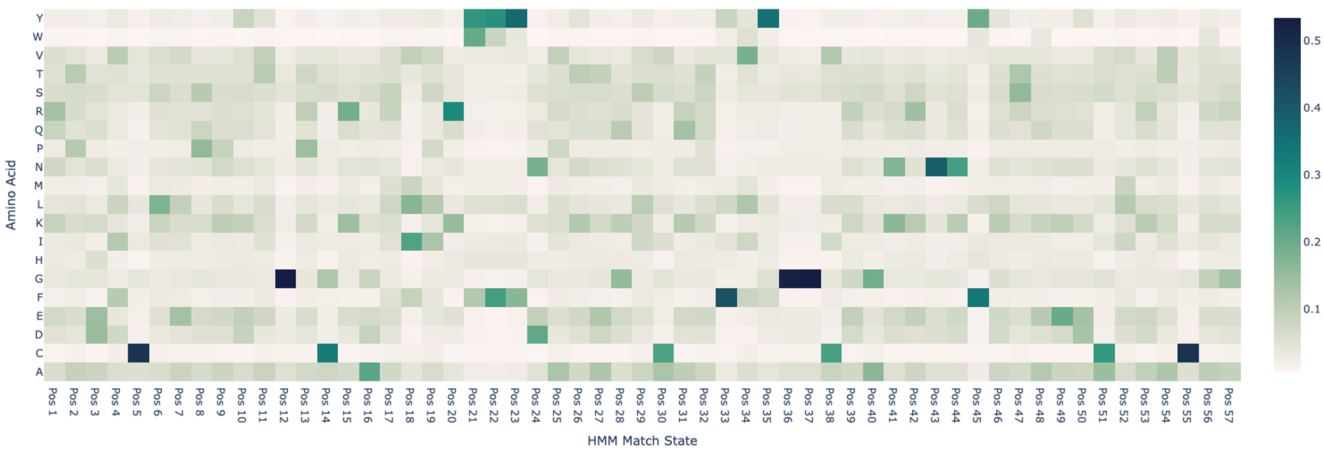

Figure 3: kunitz_clean HMM emission probabilities

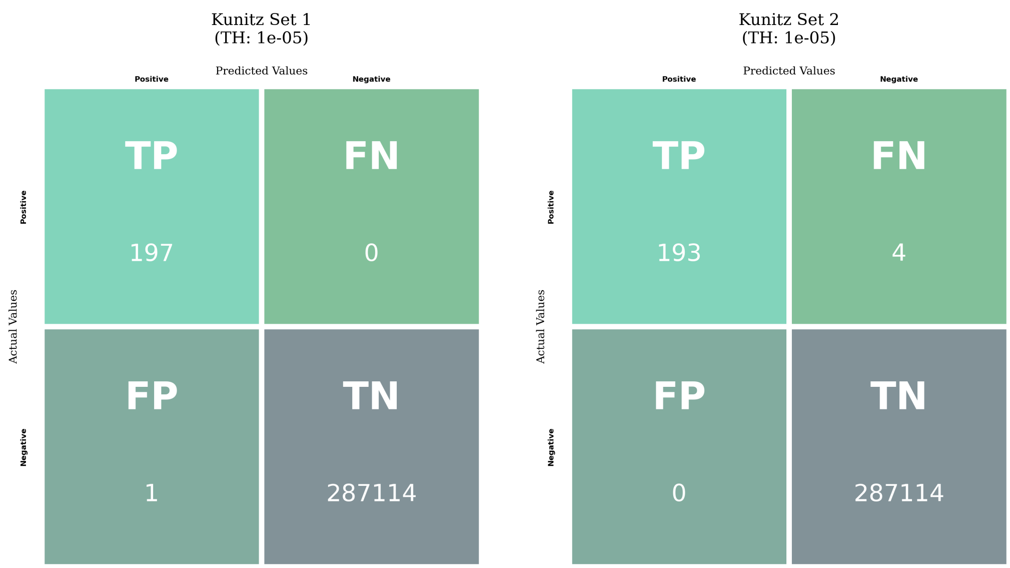

Figure 5: confusion matrix for set 1 and for Set 2

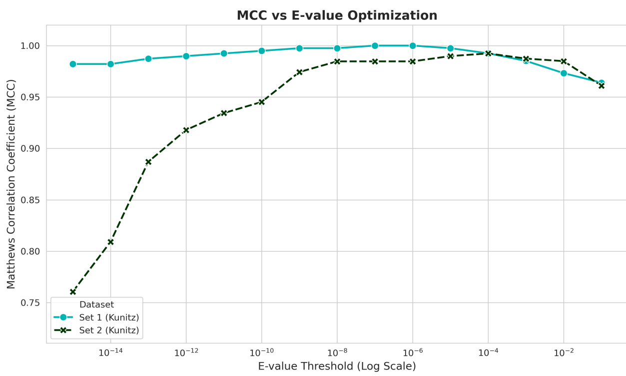

Fig 6: relationship between MCC and E-Value

4. DISCUSSION & CONCLUSION

The final aim of this project was to construct a Hidden Markov Model capable of providing reliable and sensitive annotation of the Kunitz domain. By adhering to rigorous structural filtering guidelines and performing manual curation of the input sequences, we successfully generated a model, **kunitz_clean.ali**, that captures the essential biological features of the family. The results emphasize the importance of high-quality training data in HMM generation. The initial identification of structural artifacts—specifically sequences lacking the six hallmark conserved cysteines—demonstrated that even within verified databases, outliers can introduce significant statistical noise. The removal of these outliers was validated by the dramatic increase in internal sequence identity, ensuring that the model's emission probabilities were anchored to structurally invariant residues. Performance evaluation through 2-fold cross-validation confirmed the model's robustness, achieving a near-perfect Matthews Correlation Coefficient (MCC) of 0.982 at a consensus E-value threshold of 10^{-5}. The overall predictive power is further illustrated by the ROC curve, which shows a near-ideal trajectory for both validation subsets, indicating high sensitivity maintained at extremely low false positive rates(Fig 7). While the model successfully discriminated between Kunitz and non-Kunitz sequences across a dataset of over 570,000 entries, the identification of a few false negatives, such as entries O62247 and D3GGZ8, highlights the inherent challenges in modelling distant homologs with lower sequence conservation.

In conclusion, the **`kunitz_clean.hmm`** model provides a powerful tool for protein annotation, balancing high specificity with the sensitivity required to detect the core Kunitz fold. Future refinements could involve incorporating these "extreme" biological variants into a more diverse training set or employing a multi-model approach to capture the full evolutionary breadth of the Kunitz superfamily.

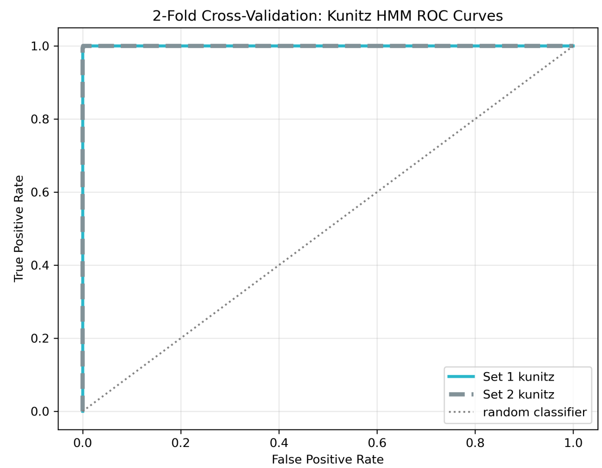

Fig 7: ROC Curve comparison between set 1 and set 2

References

[1] Grigoreva, A. A., Tereshchenkov, A. G., Savinkova, L. K., & Knorre, D. G. (2020). Kunitz-type inhibitors: A review on their action against serine proteases. *Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics*, *1868*(1), 140294.

[2[Bajaj, M. S., Ogueli, G. I., Kumar, Y., Vadivel, K., Lawson, G., Shanker, S., Schmidt, A. E., & Bajaj, S. P. (2011). Engineering Kunitz Domain 1 (KD1) of human tissue factor pathway inhibitor-2 to selectively inhibit fibrinolysis. *Journal of Biological Chemistry*, *286*(6), 4329–4340.

[3] Wettstein, L., Kirchhoff, F., & Münch, J. (2022). The Transmembrane Protease TMPRSS2 as a Therapeutic Target for COVID-19 Treatment. *International Journal of Molecular Sciences*, *23*(3), 1351.

[4] Bokova, D., Bechtel, M., McLaughlin, K. M., McGreig, J. E., Klann, K., Fullam, A., Dosari, M. E., Cinatl, J., Jr, Michaelis, M., & Wass, M. N. (2020). Aprotinin inhibits SARS-CoV-2 replication. *Cells*, *9*(11), 2377.

[5]Mirita, M., Steinegger, M., &Söding, J. (2019). MMseqs2 desktop and local web server app for fast, interactive sequence searches. *Bioinformatics*, *35*(16), 2856–2858.

[6] Dong, R., Peng, Z., Zhang, Y., & Yang, J. (2018). mTM-align: an algorithm for fast and accurate multiple protein structure alignment. *Bioinformatics*, *34*(10), 1719–1725.

[7] Durbin, R., Eddy, S. R., Krogh, A., & Mitchison, G. (1998). *Biological sequence analysis: Probabilistic models of proteins and nucleic acids*. Cambridge University Press.

[8]Wheeler, T. J., Clements, J. &Finn, R. D. (2014). Skylign: a tool for creating informative, interactive logos representing sequence alignments and profile hidden Markov models. *BMC Bioinformatics*, *15*(1).